

TrES-3b

R-0299

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_180417 · created 2026-07-17T20:08:06+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458225.944 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.196 +/- 0.056
Transit depth (Rp/Rs)^2	3.84 +/- 2.2 [%]
Orbital Inclination (inc)	86.95 +/- 2.86
Airmass coefficient 1 (a1)	1.05 +/- 0.53
Airmass coefficient 2 (a2)	-0.18 +/- 1.37
Scatter in the residuals of the lightcurve fit is	50.61 %
Best Comparison Star	None
Optimal Aperture	1.46
Optimal Annulus	4.25
Transit Duration (day)	0.08 +/- 0.01

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.196 ± 0.056 vs 0.16309 (deviation 0.48σ)
Duration fit vs published	0.08 d vs 0.0567 d (deviation 1.9σ)
Mid-time O–C	-2.7 min
Depth SNR	2.9
Residual scatter	50.61 %

Light curve

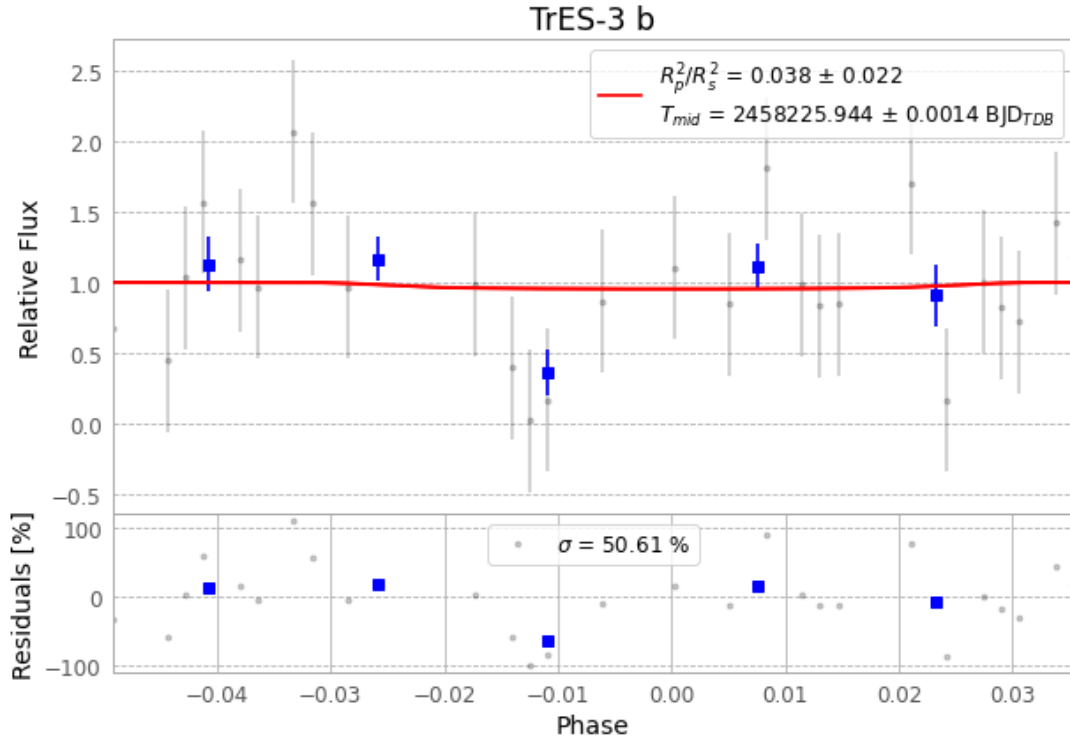


Figure 1 — FinalLightCurve_TrES-3 b_2018-04-17.png (SHA-256 5997822f7c60018a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [545, 206], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 5df2bfeb292b284a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 3612b4b

Data provenance

55 FITS frames (36 MB), TRES-3180417090312.FITS → TRES-3180417114512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-180417.log (77720bf7bfe2...), FinalLightCurve_TrES-3 b_2018-04-17.png (5997822f7c60...), FinalLightCurve_TrES-3 b_2018-04-17.csv (88d35b8ff22b...)

Compute resources

Wall time 160.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 20:08Z.